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EDU201 Midterm Preparation Mega file
BY PIN2 and MUHAMMAD
(MAS All Rounder)

Lesson no 1

INTRODUCTION TO LEARNING

Topic 1: Definition of Learning Basic

principles:

All humans learn

- Infants learn
- Children learn
- Adults learn

Learning is not uniquely human – all living things learn.

Learning is commonly defined as:

A process that brings together.

- **Cognitive** (mental, intellectual, reasoning)
- **Emotional,** and
- **Environmental** influences and experiences

Learning is commonly defined as:

For acquiring, enhancing or making changes in one's

- **Knowledge**
- **Skills**
- **Values, and**
- **Worldwide views**

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Topic 2: Nature of Learning

The depth or nature of the changes involved are likely to be different:

1. Learning as a quantitative increase in knowledge: Knowing a lot
2. Learning as memorizing: storing information to reproduce.
3. Learning as acquiring facts, skills and methods to retain and use.

4. Learning as making sense or abstracting meanings.
5. Learning as interpreting and understanding reality in a different ways.

Topic 3: Key Elements of Learning

- Learning is a product.
- Learning is a process.
- Learning as a product focuses on outcome – the end product of some process. It can be recognized or seen.
- This approach highlights a crucial aspect of learning change.
- Learning as a process focuses on what happens when the learning takes place.
- Explanations of what happens constitute learning theories.

Topic 4: Learning as a Product

- Same performance measures applies to all students
- Emphasis is on doing activities – and getting them right, because they are graded.
- Teacher thus assesses who needs more practice in writing, multiplication or something else.
- But this does not tell when the students acquires the skill, or if he/she is more advanced in it.

Topic 5: Learning as a Process

- Differentiation and individualization occur
- Students are viewed as having diverse needs
- Teacher accommodates every student's needs (learning styles)
- Worksheets, exercises, activities and even home work are individualized.
- Mistakes are considered as part of learning.
- Assessment compares student's current achievement to his/her previous level of proficiency or competency, not against the achievements of the peers.

Topic 6: How do People Learn?

Learning is simple, right?

- It is the process of moving information from out there – from a textbook, a friend's notes, Discovery channel – to in here, inside our heads and making that knowledge our own.
- But it turns out that learning is not so simple and obvious.
- Easy answer: We do not know for sure.

- Difficult answer: we have multiple theories that provide answers from many different perspectives.
- These come from psychologists, philosophers, sociologists, anthropologists, evolutionary biologists, linguists, neuroscientists….

Lesson no 2

THEORY AND LEARNING THEORIES

Topic 7: Theory and Learning Theory

Definition of Theory A

theory:

- provides a general explanation for observations made over time.
- explains and predicts behavior.
- can never be established beyond all doubt.
- may be modified.

A learning theory:

Describes how people learn.

Topic 8: Importance of Learning Theories

Key Value 1

- Gives language and a conceptual framework for understanding the examples of learning that we observe.

Key value 2

- Suggests where to look for solutions to practical problems.
  Importance of Learning theories.
- Provides better idea of purpose behind teaching and learning.
- Equips the teacher to meet the learning needs of learners effectively.
- Enhances teaching and learning by providing guidelines to different methods of teaching and use of instructional resources including technology.

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Topic 9: Broad Domains of Learning Theories

- Behaviorism
- Constructivism
- Cognitivism

We have many independent, specialized knowledge structures, rather than one cohesive knowledge structure.

Behaviorism

- It focuses on objectively observable behaviors and discounts mental activities.
- Behaviorists define learning as the acquisition of new behavior.

Cognitivism

- How we acquire, store and process information
- It looks beyond the behavior to explain brain-based learning that may have link with memory, problem solving and attention.

Constructivism

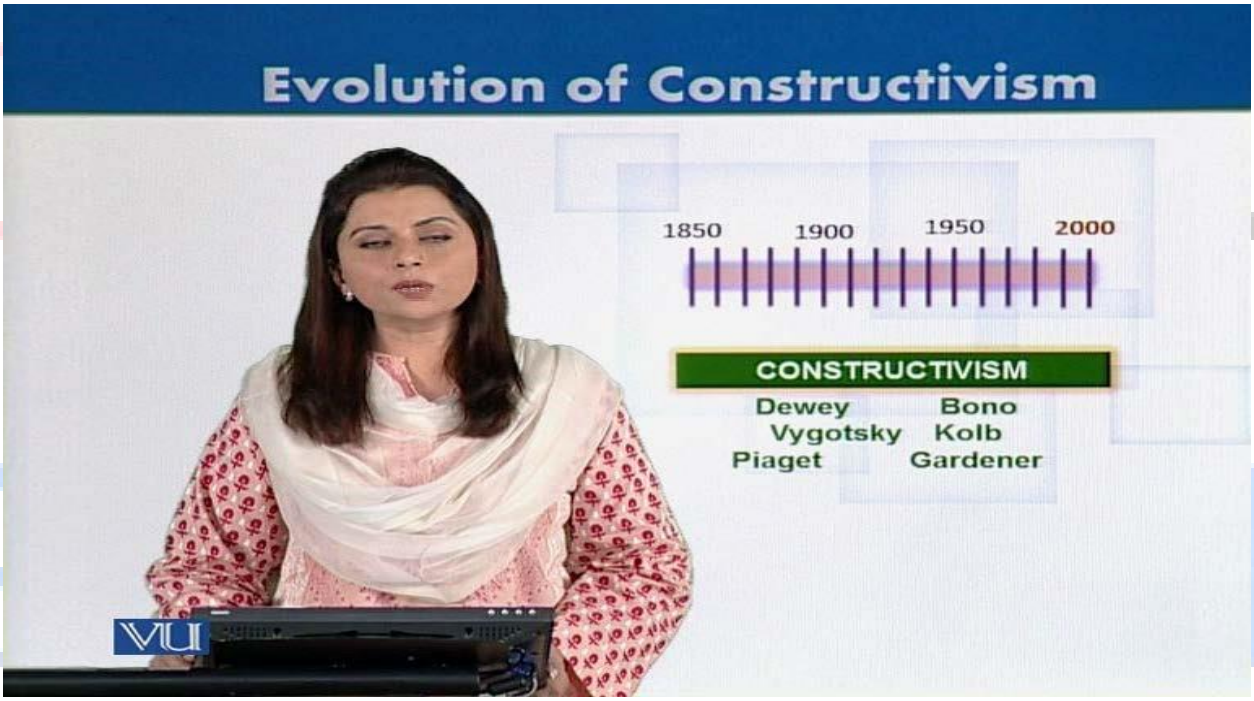
- Learning is viewed as process in which the learner actively constructs or builds new ideas or concepts based on experience/interpretation/social interaction. **Topic 10: Evolution of Behaviorism Behaviorism:**
- The development of behaviorism, the first domain psychologists carried out experiments in laboratories under strict conditions and thus observe behavior as never before.
- Behaviorism is a learning theory that concentrates only on behaviors that rae readily observable and disregards mental actions.
- It has a high impact on education, objectives/outcomes in curriculum, such as the importance of feedback, skills development and training, and pre-alignment of content, teaching methods and assessment. **Topic 11: Evolution of Cognitivism Cognitivism:**
- How we acquire, store and process information:
- Eventually behaviorism began to falter because aspects of learning such as memory, language and other metal abilities could not be considered within its core logic.

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- While behaviorists only look at the observable behaviors, cognitivists conversely consider the mental processes of the learner to be of primary importance. Cognitivism has been referred to as an "information processing" view of learning.
- Curriculum design became more flexible with ideas of continuous assessment, group based learning and applied practice.
- Application of skills rather than rote memorization is a goal in schools today. **Topic 12:**

Evolution of Constructivism Constructivism:

- During the later 1980s and the 1990s, these cognitive theories were challenged.
- Emphasis was on the importance of interactions and the sociocultural context of learning.



- A learning theory that contends that knowledge is not merely transmitted from teacher to student, but it is actively constructed in the mind of the learner out of their experiences in the world.

Lesson no 3

BEHAVIOURISM I

Topic 13: Concept – Behaviorism

- Behavioral learning theory is the study of external behavior.
- It implies that learner starts with a clean slate.
- And learners from his or her response to the stimulus in his or her environment.
- Thinking does not have to take place for a person to learn.
- Behaviorists assert that the only behaviors worthy of study are those that cab be directly observed.
- Like actions, rather than thoughts or emotions, which are the legitimate object of study.

Topic 14: Basic Assumptions

- Learning is manifested by a change in behavior.
- The environment shapes behavior.
- Principles of contiguity (how close in time two events must be for a bond to be formed)
- Reinforcement (any means of increasing the likelihood that an event will be repeated) are central to exploring the learning process.

Topic 15: The Process ○ For behaviorism, learning is the acquisition of new behavior through conditioning.

- ○ Conditioning is a process of behavior modification by which a subject comes to associate a desired behavior with a previously unrelated stimulus.
- ○ And a response becomes more frequent or more predictable in a given environment as a result of reinforcement.

There are two types of possible conditioning ○

Classical ○ **Operant**



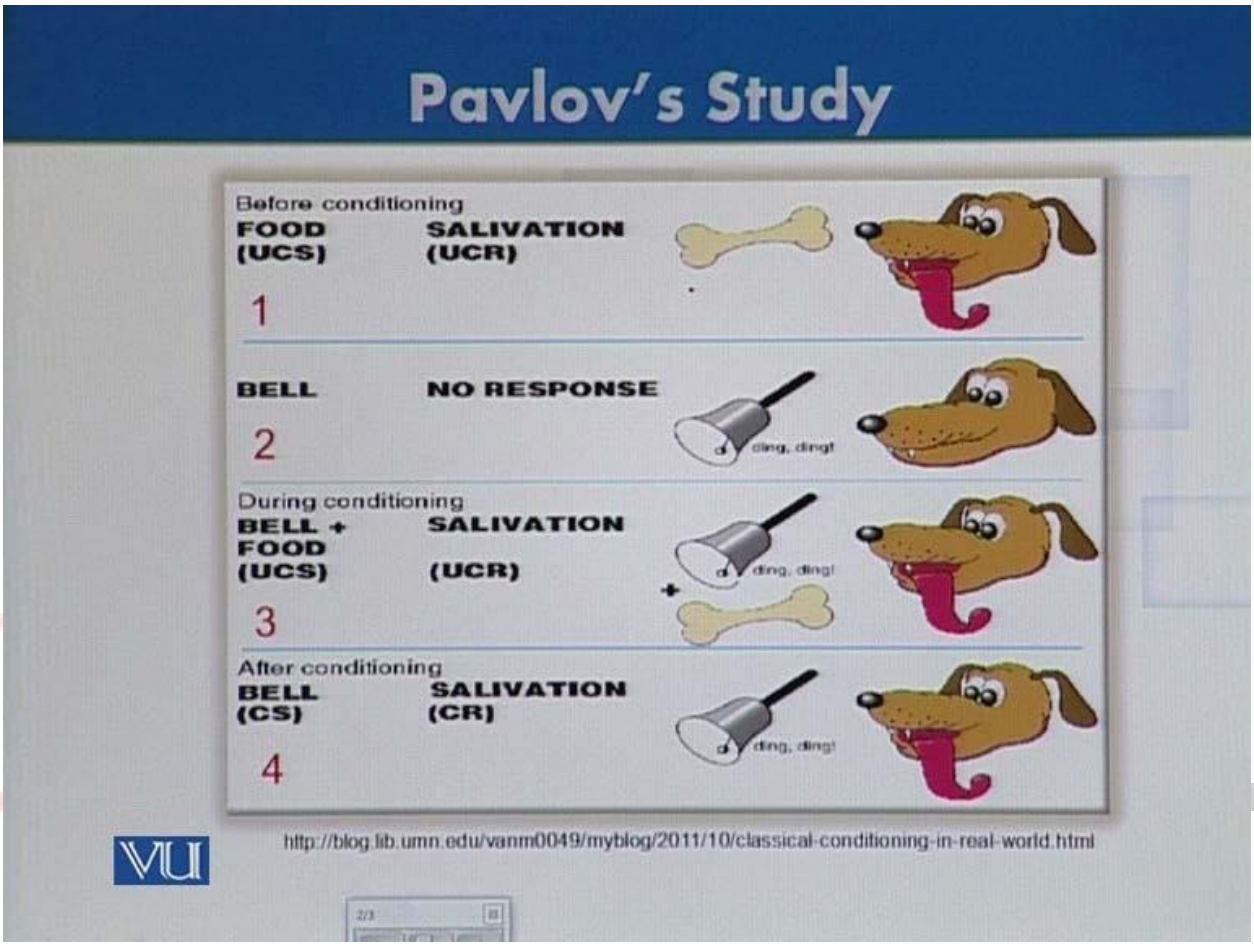
Topic 16: Classical Conditioning

Learning by association

- Natural reflex that occurs in response to a stimulus.
- A naturally occurring stimulus is placed with a neutral stimulus allowing the neutral stimulus to, in time, evoke a natural reflex.
- Our response is involuntary.
- Ivan Pavlov (1849-1936) discovered Classical Conditioning
- In his digestive research, Pavlov noticed that dogs salivated in absence of food and smell.
- This led him to another very important study.

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Topic 17: Pavlov Study



Pavlov (1902) started from the idea that there are some things that a dog does not need to learn. For example, dogs don't learn to salivate whenever they see food. This reflex is 'hard wired' into the dog. In behaviorist terms it is an unconditioned response (i.e. a stimulus-response connection that required no learning). In behaviorist terms, we write:

Unconditioned Stimulus (Food) > **Unconditioned Response** (Salivate)

Pavlov showed the existence of the unconditioned response by presenting a dog with a bowl of food and the measuring its salivary secretions (see image below).

However, when Pavlov discovered that any object or event which the dogs learnt to **associate** with food (such as the lab assistant) would trigger the same response, he realized that he had made an important scientific discovery. Accordlingly, he devoted the rest of his career to studying this type of learning.

Pavlov knew that somehow, the dogs in his lab had learned to associate food with his lab assistant. This must have been learned, because at one point the dogs did not do it, and there came a point where

they started, so their behavior had changed. A change in behavior of this type must be the result of learning.

In behaviorist terms, the lab assistant was originally a neutral stimulus. It is called neutral because it produces no response. What had happened was that the neutral stimulus (the lab assistant) had become associated with an unconditioned stimulus (food).

Topic 18: Watson and Little albert's Case Study

- An American psychologist who promoted a change in psychology of learning.
- Also conducted the controversial "Little Albert" experiment.

Albert

- 9 month's old infant.
- Was first shown a white rat; did not react
- Next shown a white rat, and heard loud noise, simultaneously.
- Done repeatedly
- By the end, whenever he saw the white rat he cried
- The infant learned to associate the rat with a scary noise, and experienced fear whenever he saw it.

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Lesson no 4

BEHAVIOURISM II

Topic 19: Operant Conditioning

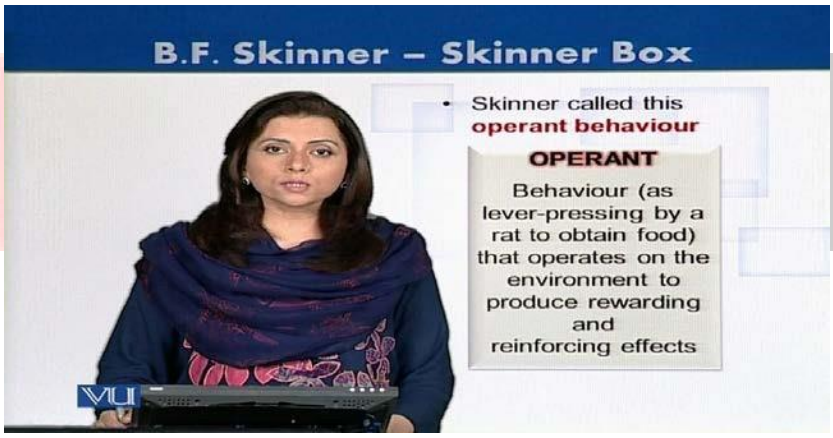
Operant conditioning (or **instrumental conditioning**) is a type of learning in which an individual's behavior is modified by its antecedents and consequences.

Operant conditioning is distinguished from *classical conditioning* (or *respondent conditioning*) in that operant conditioning deals with the reinforcement and punishment to change behavior. Operant behavior operates on the environment and is maintained by its antecedents and consequences, while classical conditioning is maintained by conditioning of reflexive (reflex) behaviors, which are elicited by antecedent conditions. Behaviors conditioned through a classical conditioning procedure are not

maintained by consequences. They both, however, form the core of behavior analysis and have grown into professional practices.

Topic 20: B.F Skinner – Skinner box

- B.F Skinner conceived the term "Operant Conditioning".
- He put a rat in a box and trained it to pull a lever to get its food – by not feeding it until it would pull the lever.
- He found that behavior did not depend on the preceding stimulus as Watson and Pavlov maintained.
- Instead, Skinner found that behaviors were dependent upon what happens after the response.





Topic 21: Edward Thorndike – Law of Effect

- Thorndike did foundational research on why consequence of behavior, such as rewards or punishments, affects our future choices.
- Devised a wooden puzzle box and put a cat in it – and placed some fish food outside.
- Measured how long cat took to get out, and eat fish – gradually, cat learned to pull correct levers.

- Concluded: All learning involves the formation of connections, which are strengthened as per the law of effect.
- And behavior that is followed by pleasant consequences is likely to be repeated - Any behavior followed by unpleasant consequences is likely to be stopped, - **Behavior modification.**

Topic 22: Classical vs Operant Conditioning

Classical: Relies on stimuli to learn.

Operant: Relies more on consequences.

Classical: Neutral stimulus that naturally and automatically triggers a response.

Operant: Requires the use of reinforcement or punishment.

Classical: Learning does not require punishment.

Operant: Punishment to make learning happen.

Passive: Learning involuntary response – classical.

Active: Learning voluntary response – operant.

Classical: Instantaneous response Dog salivating when a bell rings.

Operant: Reaction is controlled Study hard to get an A instead of F.

Topic 23: Behaviorists Paradigm of Learning

Uses in the classroom Teacher

perspective:

- Reward systems (reinforcement) to promote learning
- Positive reinforcement
- Have consequences for negative behavior
- Teachers models and students imitate **Student perspective:**

- Motivated to learn because they can get a reward
- Wants to do well because others are receiving rewards
- Criticism and Strengths

- One – dimensional approach to understanding human behavior.
- Does not account for free will and internal influences such as moods, thoughts and feeling.
- Does not account for other types of learning, especially learning that occurs without the use of reinforcement and punishment.
- Since behaviorism is based upon observable behaviors.
- It is easier to quantify and collect data and information when conducting research.

Topic 24 : Criticism and Strengths

One - dimensional approach to understanding the human behavior.

Does not account for free will and internal influences such as moods, thoughts and feelings.

Does not account for other types of learning, especially learning that occurs without the use of reinforcement and punishment.

Since behaviorism is based upon observable behaviors.

It is easier to quantify and collect data and information when conducting research,

Final thoughts

Not as dominant as was during the middle of the 20th century – stills remains an influential force.

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Lesson no 5

COGNITIVISM

Topic 25: Foundation of Cognitivism

- A theoretical framework for understanding the mind - Explores mind from the perspective of process.
- A response to behaviorism- which cognitivists said neglected to explain cognition - Behaviorists acknowledged the existence of thinking, but identified it as a behaviour.

- Cognitivist argues that the way people think impacts their behaviour and thus cannot be a behavior in and of itself.

Topic 26: Theoretical Approach

- Cognitive theories focus on the conceptualization of students' learning processes:

How information is:

- Received,
- Organized,
- Stored, and
- Retrieved by the mind

Cognitive theories are considered more appropriate for explaining complex forms of learning like reasoning, problem solving, information processing, etc.

- **Goal of Instruction:** to transfer knowledge to the students in the most efficient and effective manner possible.
- **Focus:** The design of environment to optimize that, transfer with efficient processing strategies.

Topic 27: Difference between Cognitivism & Behaviourism

Behaviourism: Observable behavior and the response of humans to environmental stimuli.

Cognitivism: Human mind and its complex processes; such as thinking, problem solving, and memory (Schunk 2004/2007b)

Cognitivism: concerns with explaining higher mental processes; perception, information processing, language and knowing.

Cognitive Theories: research based more on humans than on animals.

Cognitivist theories are often based on an information processing model.

Cognitive processes: use existing knowledge and generate new knowledge.

Topic 28: Jean Piaget

- Jean Piaget is credited to have most profoundly affected the cognitive paradigm of knowledge.

- He investigated the hidden side of children's minds
- According to Piaget, our thinking processes change radically, though slowly, from birth to maturity because we constantly strive to make sense of the world.
- "How do we do this?"
- The most important influences on the way we make sense of the world is maturation.
- Maturation's main contribution to cognitive development is in neurological growth (and the development of the endocrine system)

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Lesson no 6

PIAGET'S STAGES OF DEVELOPMENT

Topic 29: Piaget's Stages of Development

There are three basic components to Cognitive Theory:

- **Schemas:** building blocks of knowledge
- **Adaptation processes:** enabling the transition from one stage to another - **Stages of development**

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Topic 30: Piaget's 4 Stages of Cognitive Development-1

Sensorimotor: Birth-2years old

- Identifies objective performance: The objective still exists when out of sight.
- Recognition of ability to control objects and acts intentionally.
- Deals with reality in terms of sensation and motor movements.

Topic 31: Piaget's 4 Stages of Cognitive Development-2

Preoperational: 2-7 years old

- Begins to use language
- Egocentric thinking: difficulty seeing things from other viewpoints - Classifies objects by single features – example, color.

- Develops capability of symbolic thought- however thinking still quite different from adults.

Topic 32: Piaget's 4 Stages of Cognitive Development-3

Concrete Operational: 7-11 years old

- During middle childhood, the child has the ability to reason like an adult in every way…
- … except for reasoning about abstract concepts e.g. justice, infinity, or the meaning of life.
- Recognizes conservation of numbers mass and weight.
- Classifies objects by several features and can place them in order

Topic 33: Piaget's 4 Stages of Cognitive Development-4

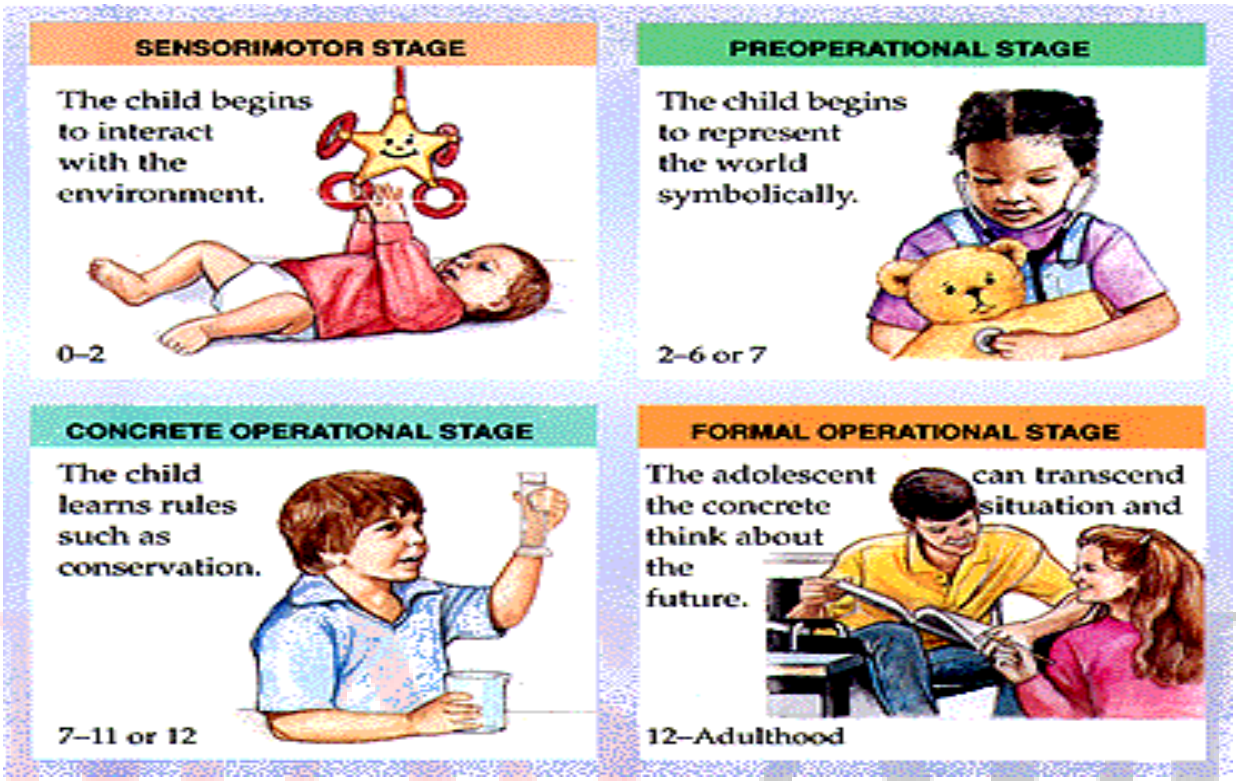
Formal Operational: 11 years and up

- By the end of childhood, most individuals have progressed to full adult cognition…
- … including the ability to reason using abstract concepts
- Concerned with the hypothetical and the future

Final Thought – Piaget

A child who had not completed certain developmental stages could not learn things from higher developmental stages.

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Topic 34: Allan Paivio Dual Coding Theory

He proposed that presenting information both visually and verbally enhances recognition and recall. We have two keys of processing information:

1. **Visual Channel**
2. **Verbal Channel**

- When both visual and verbal representations are used, people both process and remember the information more effectively.
- Paivio's work has implications in many areas including human factors, interface design as well as the development of educational materials. **Topic 35: Uses in classroom Teacher's perspective:**
- Plan curriculum based on what students already know and what they should learn
- Adjust instruction and assessment based on students' learning styles – use visuals
- Develop knowledge and skill from simple to complex (hierarchical learning)

Topic 36: Criticism of Piaget's Theory

- **Nature of Development:** Piaget's theory does not explain why development from stage to stage occurs.

- **Individual differences:** Theory ignores that some individuals move from stage to stage faster than others.
- **Nature of sages:** The functioning of a person at a given age may be so variable from domain to domain that it is not possible to place the person in a single stage.
- **Underestimating children's abilities:** According to many researches preschool children know much more about the concept of number than Piaget thought.
- **Cognitive development and culture:** One final criticism of Piaget's theory is that it overlooks the important effects of the child's cultural and social group.

Lesson no 7

CONSTRUCTIVISM I

Topic 37: Introduction

What each person 'sees' depends more on what is already stored in that person's brain. A person constructs his/her own knowledge. That is- individuals construct knowledge for themselves – each learner individually constructs meaning – as he or she learns.

Topic 38: Core Idea

- People construct their own understanding and knowledge of the world, though experiencing things and reflecting on those experiences. Individuals "construct" meaning from the world around them.
- When we come upon something new, we join it with our previous ideas and experience.
- Either we change our old view or belief.
- Or discard the new information

Think about how you learn? When you hear, read, or see something new, does it help to talk about it and reflect on it to better understand the new information?

Topic 39: Evolution of Constructivism

- The concept of constructivism has roots in classical antiquity.

- Goes back to Socrates's dialogue – asked his followers directed questions that led students to realize for themselves the weaknesses in their thinking.
- Socrates claimed that basic conditions for learning exist in the cognition of the individual (Kanuka & Anderson, 1998)
- It was Piaget's theory of intellectual growth that had the primary influence on the development of current positions.
- Specifically, Piaget first emphasized the processes of conceptual change as interactions between existing cognitive structures and new experiences.
- In recent times, John Dewey and Jean Piaget developed theories of childhood development and education (what we now call progressive education) that led to the evolution of constructivism.

Topic 40: Theoretical Framework – Social Constructivism

Two schools of thought fall within the Constructivist view:

- Social Constructivism
- Cognitive constructivism
- **Lev Vygotsky** discovered how meanings and understandings grow out of social encounters.
- **Zone of Proximal development:** children, working on their own rarely do as well, as when they work in collaboration with an adult.
- The process of engagement with the adult enables them to refine their thinking or their performance (the more knowledgeable other MKO) - Social interaction precedes development.

Topic 41: Cognitive Constructivism

Jean Piaget discovered how the individual learner understands things, in terms of developmental stages and learning styles. There are two components:

1. **Ages and stages:** predicts what children can and cannot understand at different ages.
2. **Theory of development:** describes how learners develop cognitive abilities.

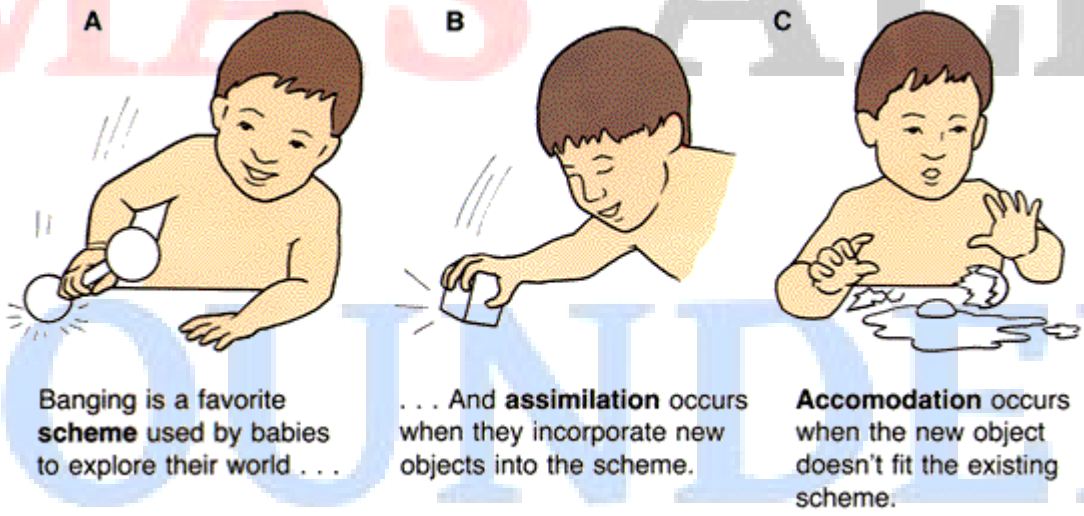
Theory of Development:

- Humans cannot be given information, which they immediately understand and use.
- Learners must construct their own knowledge, they build their knowledge through experience.
- Experiences enable them to create schemas – metal models of the world.

- These schemas are changed, enlarged, and made more sophisticated through two complementary processes: assimilation and accommodation.

Topic 42: Assimilation and Accommodation **Assimilation:**

- The individual cognitively adapts to and organizes information by using current, preexisting schemas to interpret experiences and the external world. - Ongoing process **Accommodation:**
- Changes in behavior and thinking take place when new information no longer fits into old ways of understanding (the old schemas)
- Existing schemas change to accommodate new information or new schemas are created that contain the new information.
- Intellectual capacities become re-shaped and recognized as the child attempts to adjust. - Thinking becomes more sophisticated.



Topic 43: John Dewey

John Dewey's Philosophy:

- Education starts with the needs and interests of the child.
- Employs project method or group learning.
- Depends heavily on experiential learning.
- Ideas are not separate from social conditions.
- Children are active, organic begins… needing both freedom and responsibility. - Allow the child to participate in planning her course of study.

Dewey's Role for Teacher:

- Not the authoritarian but the facilitator.
- Encourages, offers suggestions, questions and helps plan and implement courses of study.
- Has command of several disciplines.

Lesson no 8

CONSTRUCTIVISM II

Topic 44: Maria Montessori Children

teach themselves:

This simple but profound truth inspired Montessori's lifelong pursuit of furthering the selfcreating process of the child.

Environment:

- Child sized equipment
- Precise organization
- Individual work spaces for each child
- Loosely arranged into centers **Materials:**
- Concept development
- Graduated difficulty/Complexity
- Self-correcting
- Sensory exploration **Teacher's role:**
- Carefully prepare the environment
- Attitude of humility
- Respect for the child's individuality **Topic 45: Learning how to learn?**

Constructivism helps student learn HOW TO LEARN

- Constructivist teachers pose questions and problems
- Then guide students to help them find their own answers.

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They may:

- Prompt students to formulate their own questions (inquiry)
- Allow multiple interpretations and expressions of learning (multiple intelligences)
- Encourage group work (Collaborative learning) **Topic 46: Classroom Implications-1**

Learning is constructed:

- Students are not blank slates upon which knowledge is imprinted.
- They come to learning situations with already formulated knowledge, ideas, and constructs – the raw material for new knowledge they will create.

Learning is active:

- Student is the person who creates new understanding for him/herself.
- Teacher facilitates and allows the students room to experiment, ask questions, try things that do not work (fear of failure is not an obstacle) **Learning is reflective:**
- Student control their own learning processes, and they lead the way by reflecting on their experiences.
- This process makes them experts of their own learning. **Topic 47: Classroom Implications-2 Learning is collaborative:**
- Students learn about learning not only from themselves, but also from their peers.
- When students review and reflect on their learning together, they can pick up strategies from one another.

Learning is inquiry based:

- Main activity in constructivist classroom is solving problems
- Students use inquiry methods to ask questions, investigate a topic, and use a variety of resources to find solutions and answers.

Learning is evolving:

- Students have ideas that they may later see were invalid, incorrect, or insufficient to explain new experiences.
- These ideas are temporary steps in the integration of knowledge.

Topic 48

Critical Perspective

- Time consuming
- Expensive
- Inefficient

Mix of other theories, no one clear theory too subjective

Learner must have a higher level of self-management/maturity of successful

Unpredictable outcomes

Encourages diversity of thought where conformity is required

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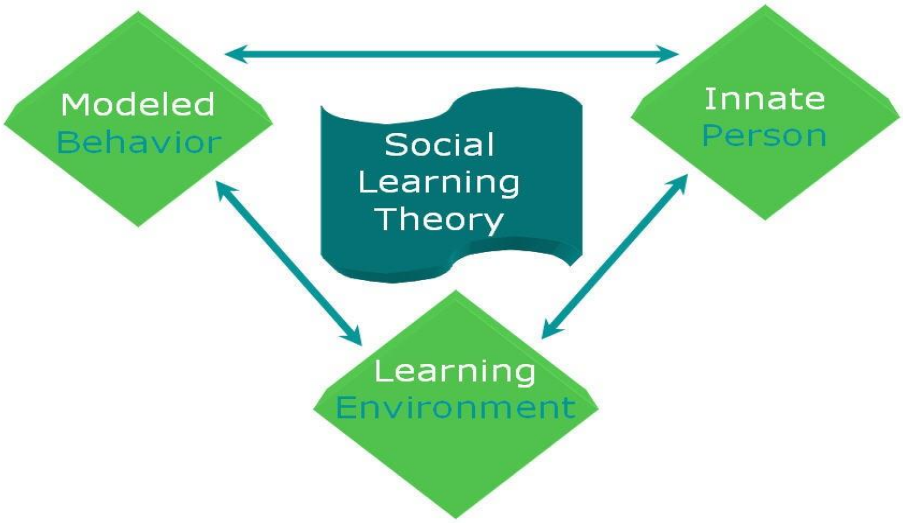
SOCIO CULTURAL THEORY

Topic 49: Foundation

- What separates humans from animals?
- What separates advanced societies from primitive societies?
- What separates advanced cognition from basic cognition?
- That is what makes us smart?

Vygotsky opens our eyes to the powerful role of culture and community in learning.

His theory presents the radical idea that our very thought and intelligence is really not our own. It's the product of history and culture.



Foundation

a) cognitive development is limited to a certain range at any given age
b) full cognitive development requires social interaction

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Topic 50: Learning as a Process of Enculturation Activity

Activity

Think about a group you were a part of in high school.

What were some of the defining qualities of this group?

1 What "look" did you need to have?
2 How did you need to talk?

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Individual and social constructivism

Individual

Individual construct meanings out of what they already know and through their interactions with the environment. **Social**

Groups or cultures construct meanings together out of what the group or culture already knows and experiences.

Vygotsky's extension

- Individuals construct meaning through their interaction with others
- That is they internalize the meaning constructed by the group or culture as they become encultured.

Topic 51: Basic Concepts

- Cultures create mental tools which transform our mental work just like physical tools transform our physical work
- As we internalize these tools we became smarter
- Language is the mother of all mental tools
- We internalize these tools as we work in our zone of proximal development

Topic 52: Social Interaction for Cognitive Development 1

- Engagement between the teacher and students
- Physical space and arrangement in learning environment
- Meaningful instruction in small or whole groups
- Scaffolding/reciprocal teaching strategies

Lesson contention

- Create lessons that engage students' interest and give them a basis for language when socially interacting
- Use technology or hands on activity to further engage them in learning
- Teaching in small groups individualized instruction meaningful and prompt feedback and better evaluation of students' feedback
- Whole class discussion allows vetting of ideas and surface misconceptions that can be addressed mid-project.

Topic 53: Social Interaction of Cognitive Development 2

- Teachers provide scaffolds so that the learner can accomplish certain tasks they would otherwise not be able to accomplish on their own
- The goal of the educator is for the student to become an independent learner and problem solver

Topic 54: Sociocultural vs Constructivism

- Both emerged as theories of knowledge in response to behaviorism
- Both are concerned with the activities that children engage in to learn

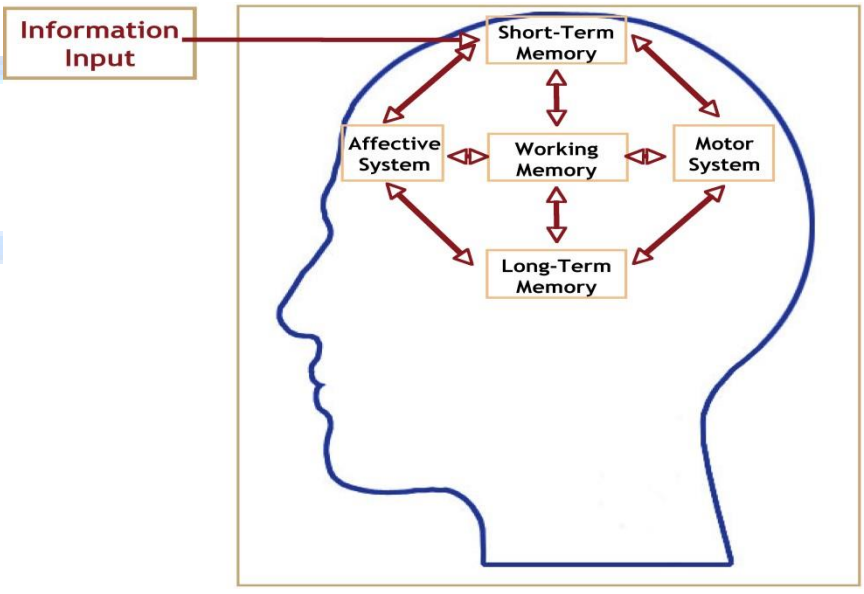
Lesson no 10

SCHEMA THEORY OF LEARNING

Topic 55: What is Schema?

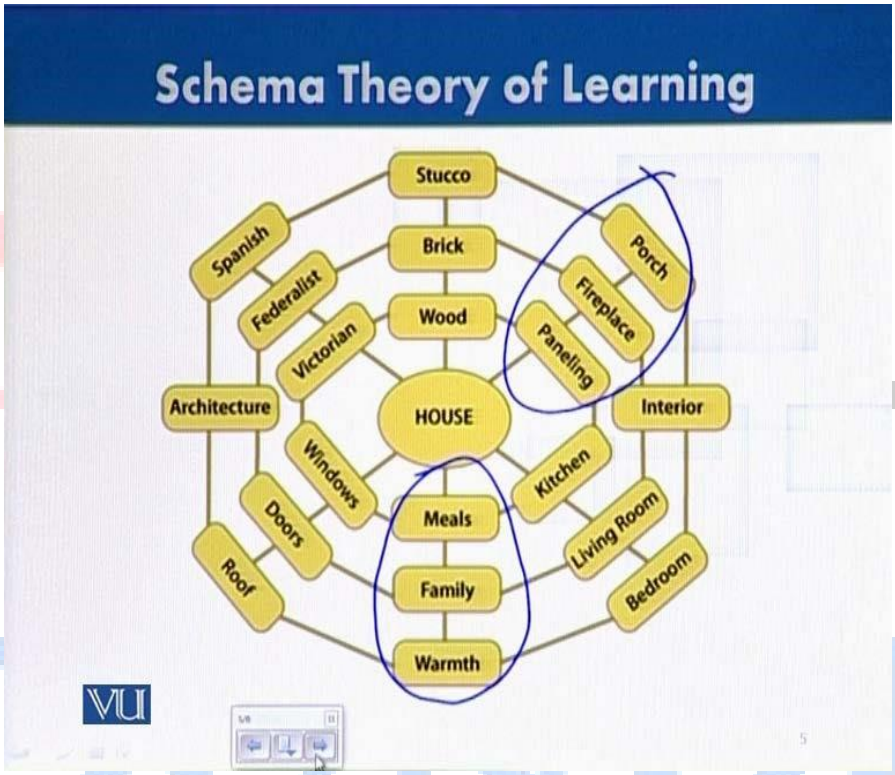
"A cognitive framework or concept that helps organize and interpret information".

In the psychology of learning, schemas refer to how you develop what you know about a topic or concept and how you integrate new information



Topic 56: Schema Theory of Learning

- The starting assumption of this theory is that " very act of comprehension involves one's knowledge of the world"
- Knowledge is a network of mental frames or cognitive constructs called schema (pl. schemata) stored in the long term memory.
- All of our generic knowledge is embedded in schemata.
- Our schemata are our knowledge.
- The schemata a person already possesses are a principal determine of what new will be learned.



Topic 57: Practical Meaning of Schema Theory 1

- Schema theory emphasizes importance of generic knowledge and concepts that will help form schemata.
- Due to the importance of prior knowledge, teachers should make sure that students have it before launching new.

Wrong question to ask

"What is it that the student does not know, and how can I get that into his head?

Right question to ask

"What is it that the student does know, and how can I use that existing knowledge as a foundation to help the child deal with new concepts that he will encounter?

Topic 58: Practical Meaning of Schema Theory 2

Motor learning

Developing a motor schema has resulted in better performance in children when learning a motor task.

- Stuttering
- Fluency-shaping
- Down-syndrome
- Motor control
- Sports

Reading comprehension ○ Schema theory is often used to assist second language learning.

○ Failure to activate adequate schema when reading a text has resulted in bad comprehension.

Mathematical problem solving

Research showed that students taught to using schemata to solve mathematical problems formulated in words performed better than their peers who were taught to solve them in four steps.

Topic 59: Problems with Schemas

- Schemata also expand and change in time, due to acquisition of new information.
- But deeply installed schemata are inert and slow in changing.
- This could provide an explanation why some people live with incorrect beliefs rather than changing them.
- New information is either assimilated or related schemata will be changed (accommodation).

Change is resisted.

- As such, sometimes an existing schema can actually hinder the learning of new information.
- **Prejudice** is one example of schema that prevents people from seeing the world as it really is and inhibits them from taking in new information.
- Existing schema may cause people to interpret situations incorrectly.

- When an event happens that challenges these existing beliefs, people may come up with alternative explanations that uphold and support their existing schema instead of adapting or changing their beliefs.

Topic 60: Criticism

- Explanations of structures of knowledge have been criticized for being rather unclear about what exactly can count as a schema and what does a schema include.
- The idea of schemata as more complex constructs of memory has also been questioned.
- Schemata as such are just networks of interacting simple (low-level) units activated at the same time.

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Lesson no 11

INFORMATION PROCESSING THEORY I

Topic 61: Introduction

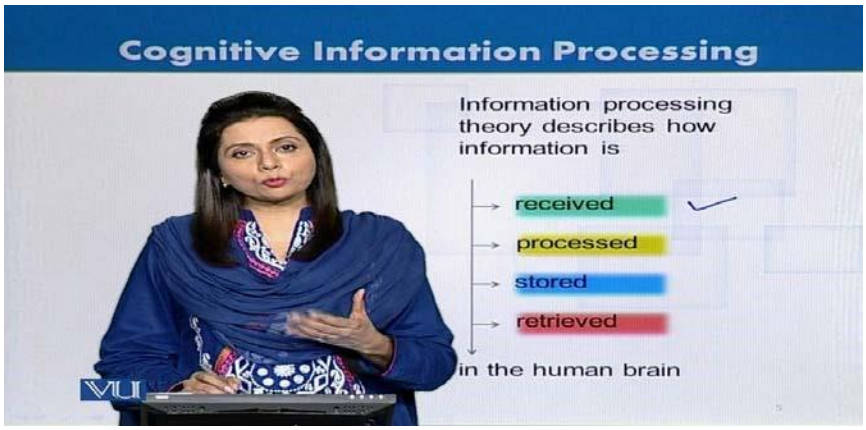
ROUNDER

Without looking at your watch, try to describe it:

- What color is the face of your watch?
- What marks the hours?
- What color are the hands?

Basic assumption of human information processing:

- Humans process the information they receive, rather than simply responding to stimuli.



Topic 62: Conceptual Framework 1

Thinking

The process of thinking includes the activities of:

- Perception of external stimuli.
- Encoding it, and
- Storing the perceived and encoded data in one's brain.

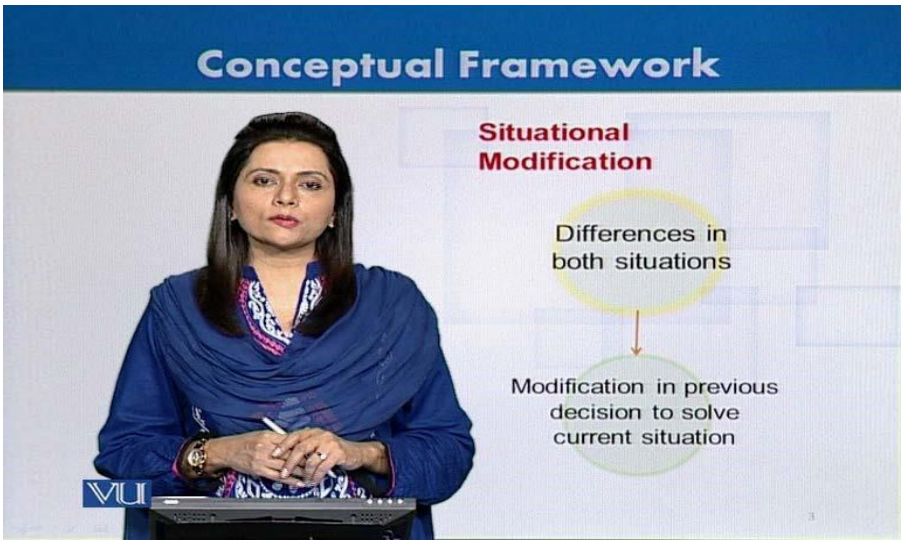
Analysis of stimuli

- This is the process wherein the encoded stimuli are altered to suit the brain's cognition and interpretation process for decision making.

Topic 63: Conceptual Framework 2

Situational modification

This is the process by which an individual uses his experience, (stored memories) to handle a similar situation existing now.



Obstacle evaluation

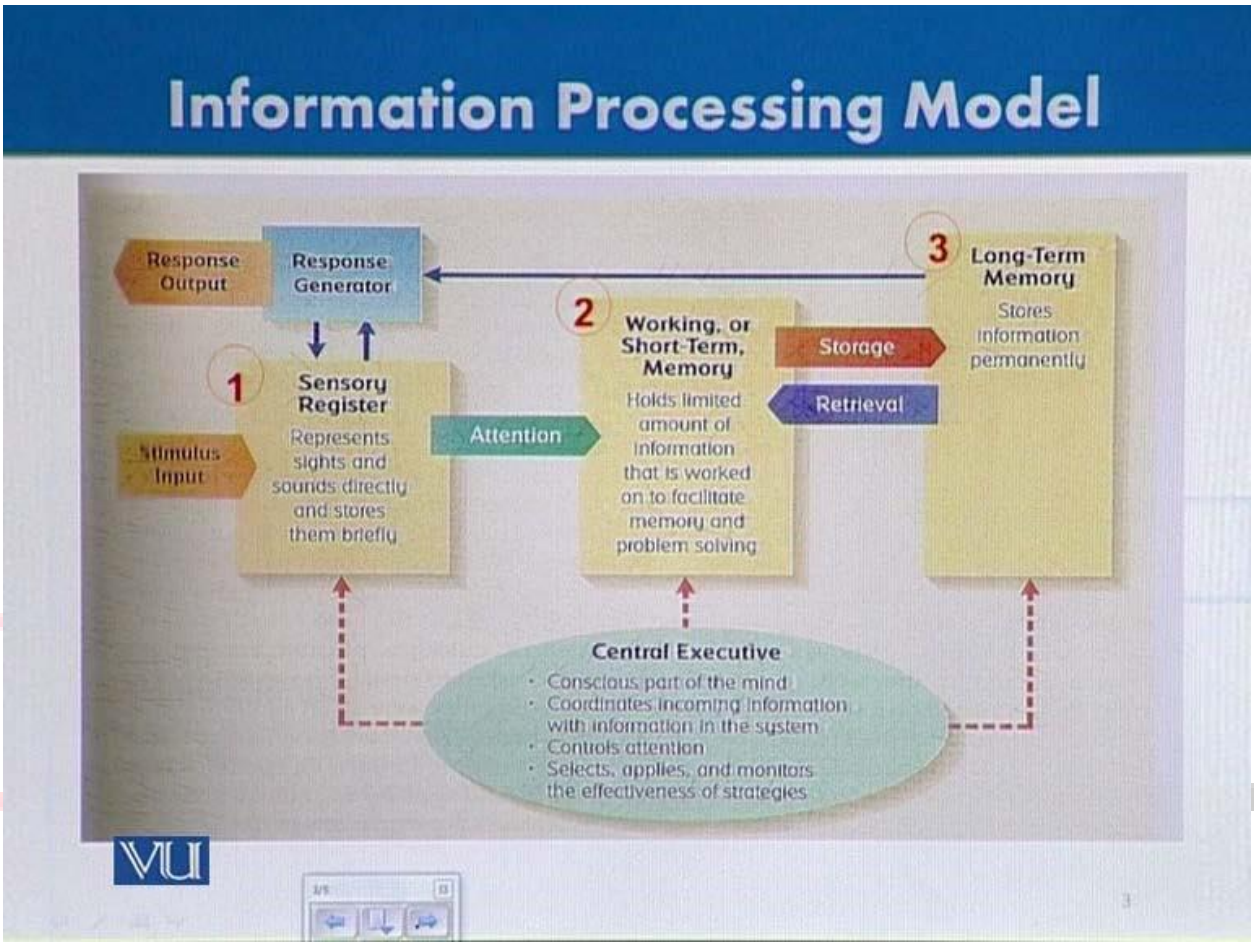
While evaluating the subject's intellectual, problem solving and cognitive level:

- Individual's development level. - Nature of obstacle or problem. **Topic 64: Information Processing Model Structure:**

Major components

1) Sensory register
2) Short-term memory (working memory) 3) Long term memory

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1. Sensory Register

- Each sensory system has its own sensory store, or sensory register, which receives and briefly holds all of the external and internal stimuli that it receives.
- The sensory register stores sensory information long enough for unconscious processes to analyze it to determine whether the input should be brought into working memory, or discarded.

Topic 65

2. Short term/Working memory

- It is the center of conscious thought (like the central processing unit CPU of a computer) where information from long-term memory and the environment is combined to help solve problems.

Topic 66

3. Long term memory

- It is the stored representation of all that a person knows.

- Items stored in long term memory lie “asleep” until they are called back into the working memory and thus put to use.
- It has two components
- Explicit memory: information you have to consciously work to remember.
- Implicit memory: information remembered by you unconsciously and effortlessly.
- Can last as little as a few days or as long as decades.
- Subject to fading in the natural forgetting process.
- Maintenance rehearsal (several recalls/retrievals of memory) may be needed to preserve long term memories.

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Lesson no 12

INFORMATION PROCESSING THEEORY II

Topic 67: Ways to Organize Information

The Central Executive

ROUNDER

- The most important component
- Responsible for monitoring and coordinating the operation of the sub-systems and relating them to Long term memory (LTM).
- For example, two activities sometimes come into conflict such as driving a car and talking. Rather than hitting a cyclist who is wobbling all over the road, it is preferable to stop talking and concentrate on driving.
- The central executive directs attention and gives priority to particular activities.

Topic 68: The Role of Attention

- The process by which people select some of the environmental input they receive for further cognitive processing.

Factors influencing attention

- Size
- Intensity
- Novelty
- Incongruity
- Emotion
- Personal significance
- Competition between similar tasks

Topic 69: Getting to the LTM

- There are two ways to move short term memory to long term memory.
- Rote learning and learning through understanding.

Rote Learning

- Learning through repetition.
- Mechanical and requires little understanding (e.g. learning multiplication tables)

Learning through Understanding

- Learning and remembering by understanding the relationships among ideas and information.
- Both types of learning and memory are useful and often are used together.

Topic 70: The Keys to Remembering

Choose to remember:

- Be interested
- Pay attention
- Want to learn and know
- What you want is an important part of learning.

Visualize in your mind what you wish to remember:

- For each major concept that you want to remember, create a mental picture and then look at it carefully for a few seconds.

Repeat what you wish to learn until you overlearn it:

- Say it in your own words. Even though you have already learned something, go over it one more time.

Topic 71: Reasons Why We Forget-1

Fading

- A memory trace is created every time a new theory is formed.
- These memory traces begin to fade and disappear as time passes.

Remembering

- Remembering what you have heard in lectures is even more difficult to recall because you are not able to slow down, pause, reflect or to reread unless you take excellent notes.
- In a study on recall after listening to a seminar, students forgot more than 90% of the points from the lecture after 14 days!

The conclusions to be made from these studies?

- Without review, most information will be lost from memory.
- The best time to review materials is within a day or two after the material has been read or presented in lecture.

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Topic 72: Reasons Why We Forget-2

Retrieval

- Sometimes a forgotten fact has not faded but misplaced in the “file cabinet” of your mind – and lost because it is forgotten.

Interferences

- As you keep adding new information, a conflict develops between the old and new information over the space available.

Interactive interferences

- When you are learning a great deal of information at one time, you tend to remember best what is read or presented first and last. The rest gets lost in the shuffle.

Lesson no 13

SITUATED COGNITION I

Topic 73: Situated Cognition

Situated cognition is a theory that posits that knowing is inseparable from doing by arguing that all knowledge is situated in activity bound to social, cultural and physical contexts

Topic 74: Basic Idea and Perspective

Theory

- Knowing is attached to doing.
- Argument
- All knowledge is situated in activity that occurs in its social, cultural and physical context.
- This model of knowledge and learning requires …… thinking on the fly …… rather than the storage and retrieval of conceptual knowledge.

The theory ….

- Emphasis perception.
- And proposes that memory plays a significantly diminished role in the learning process.

Topic 75: Background

Background: By the 1970s……

- Growing dissatisfaction among the school graduates' inability to transfer skills learned in school to the workplace e.g. Math.
- Some attempts were made to connect the syllabus with the real world. - **Problems of determining the real context.**

Topic 76: Development

Development: Allan Collins 1988

Defines situated learning as:

- The notion of learning knowledge and skills in contexts that reflect the way they will be used in the real life.
- This concept encourages educators to place their students in an environment that is as similar as possible to the context in which their learning will actually be used.



Topic 77: Situated Cognition Theory

- Situated Cognition produce knowledge through activity.
- Knowledge is a product of
  - Activity
  - Context
  - Culture

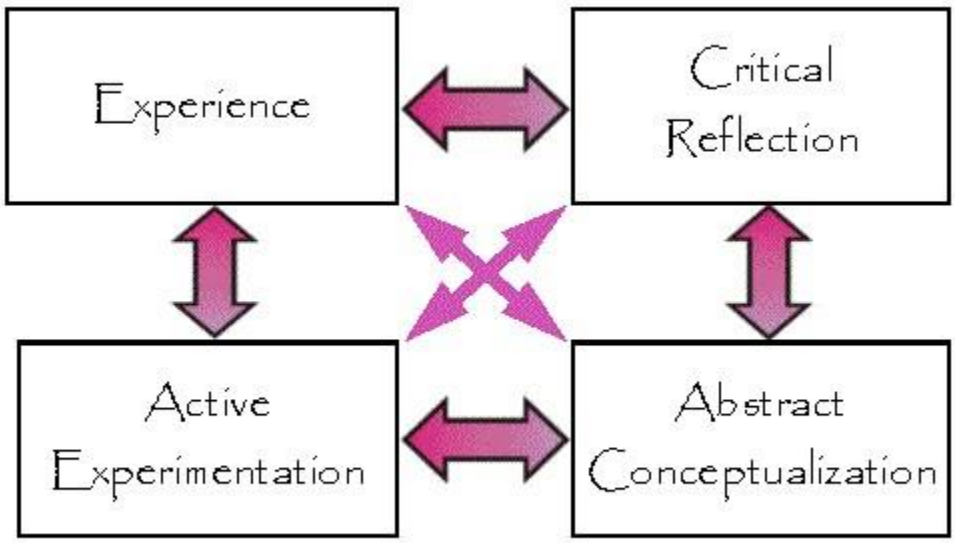
Community of Practice

- **Embedded learning in activity.** - **Use a social and physical context** - **Learning is:**

  - Demand driven
  - A social act
  - Creates individual identity formation

Community and Apprentice

- **Learners perceive**
- Implicit knowledge
- Explicit knowledge
- **By observing experts**
- Educational
- Workplace
- Environment



ROUNDER

Topic 78: Traditional Education vs Situated Cognition

Brown Collins and Duguid say……

Traditional education

- Separation between "knowledge" and "doing".
- Knowledge treated as integral, theoretically independent and self sufficient.

Knowledge Situated and Evolving through Activity.

Situated Learning Environments Four

concepts:

○ Context (environment, setting) ○ Content

(concepts, activities, situations) ○ Facilitation

(internalize, information) ○ Assessment

(cognitive growth)

Lesson no 14

SITUATED COGNITION II

Topic 79: Learning Communities and Instructor's Role

Learning communities

Build upon:

- Situatedness
  Embedded learning in rich social situations (context)
- Commonality
  Create a sense of enculturation and identity
- Interdependency
  Share expertise and perspective
- Infrastructure
  Mechanism for accountability through structure.

Motivation

- The situative view emphasizes how people's identities are formed by their participation in a group.
- According to this way of thinking, students can be motivated to learn by participating in communities where learning is valued.

Instructor's role

- Move away from providing and structuring information.
- Focus on modeling, coaching and scaffolding.
- Create knowledge to solve contextual real-life problems.

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Topic 80: Classroom Implications

- Knowledge is not an object and memory is not a location; knowledge is sociallyconstructed.

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- Knowledge is thus located in the actions of people and groups of people; it evolves naturally as a result of individuals participating in and negotiating their way through new situations.
- Mediation artifacts: cultural models are not held by individuals but live in the practices of a community and how people engage with each other, as well as any tools they use and the specific cultural context.
- As situations shape individual cognition, individual thinking and action shape the situation.
- The reciprocal influence signifies concept of systematic causality in place of more commonly assumed liner causality. (Wilson and Myers, 2000)

Topic 81: Situated Cognition in Education

- Knowing how to participate in social practices has a large role in schools.
- Much of students' identities are formed by the groups they participate in at school, e.g. sports, debates, journalism, music.
- A push for "authentic learning" where students learn content matter that is situated in areal world context.
- Students are encouraged to think and work like scientists, historians, mathematicians, etc.

Topic 82: Phenomenon Explained by the Theory

There is a mismatch between the learning situation in school and the real world situations.

- Learning is conceived as increasing participation in communities of practices.
- Learning is a co-constitutive process in which all participants change and are transformed through their actions and relations.

There is a failure of knowledge to transfer.

- Knowledge accrues through the lived practices of the people in the society.
- Knowledge remains inert and unused if taught in contexts that separate knowing from doing.

Training by abstraction is of little use.

Learning involves social participation.

- Hence, cognition takes place within the world and not in minds.

-

Learning is inherently a social phenomena.

- Cognition is a matter of sign activity.
- Or semiosis i.e. the continuously dynamic and productive activity of signs.

Topic 83: Benefits of Situated Cognition for Students

1. Students learn about the cognitions for applying knowledge.
2. Students are more likely to engage in invention and problem solving.
3. Students can see the implications of knowledge.
4. Students are supported in structuring knowledge in ways appropriate for later use.

Topic 84: Critiques of Situated Cognition

- While the theory takes into account the social, the individual tends to be ignored.
- Transfer is problematic since knowledge does not transfer between the tasks.
- The theory can not account for learning through abstraction or generalization (most school learning).

Theory's claim: Knowledge can not be separated from the context of its use.

- But this is a demonstration that skills practiced out side of schools do not generalize to schools.

Theory's claim: Transfer is difficult since knowledge does not transfer between the tasks.

- Objection: The psychological literature contains both success and failures to achieve transfer.

Theory's claim: Training in abstraction is of little use.

- Objection: research in psychology shows training is often more effective when nearly independent parts are practiced first, before combining them.

Theory's claim: Training in abstraction is of little use.

In team sports and orchestras, more time is spent on individual practice than group practice, although both are necessary.

-

Lesson no 1

MASLOW'S NEED HIERARCHY I

Topic 85: Humanistic Learning Theory

Introduction

Humanistic approach:

- The approach emphasizes individual's natural motivation towards creativity and self actualization.

Self actualization

- According to humanistic psychologists, we are motivated not merely to survive, but to become better and better.
- This process is called self actualization.
- Personal growth and fulfillment moving towards bigger goal.

According to Huitt (2001) people act with intentionality and values.

This is in contrast with……

- Behaviorists view of "Operant Conditioning" which argues that all behavior is the result of the application of consequences.

And…..

- Cognitive belief that constructing meaning or discovering, knowledge is central to learning.

The fundamental humanistic beliefs:

- That people are naturally good.
- That mental and social problems result from deviations from this natural tendency.
- And that people have the ability to shape their own destiny, and this is not driven by biological instinctive influences.
- Humanistic focus on human freedom, dignity and potential.

Topic 86: Origins

- Humanistic education has its roots in Renaissance philosophers (period roughly between 1350 and 1650) who emphasized the study of the humanities (academic disciplines that study human culture)
- These in turn became the basis for Classical models of Education.
- According to George Makdisi, certain aspects of Renaissance humanism are rooted in medieval Islamic world.
- A wide range of Islamic writings on poetry, history and philosophical theology show that medieval Islamic thought was open to the Humanistic ideas of "Individualism and Liberalism".

Humanistic approach became popular during 1950s and 1960 after the WWII and the cold war.

- During this time period, psychologists were seeking a more optimistic and nurturing approach in response to the stresses that came with the war.
- Carl Rogers and Abraham Maslow are most closely associated with the Humanistic psychology of learning.

Topic 87: Abraham Maslow (1908-1970)

- Considered to be the founder of Humanistic psychology.
- Developed the Hierarchy of Needs, theory that remains valid for understanding human motivation, training and personal development.
- The hierarchy of needs illustrate his theory that people's behaviors are guided by a sequence of needs.
- Maslow argued that humans possess unique qualities that enable them to make independent choices, thus giving them control of their destiny.

Maslow's need hierarchy theory:

- People have hierarchy of increasingly higher level needs.

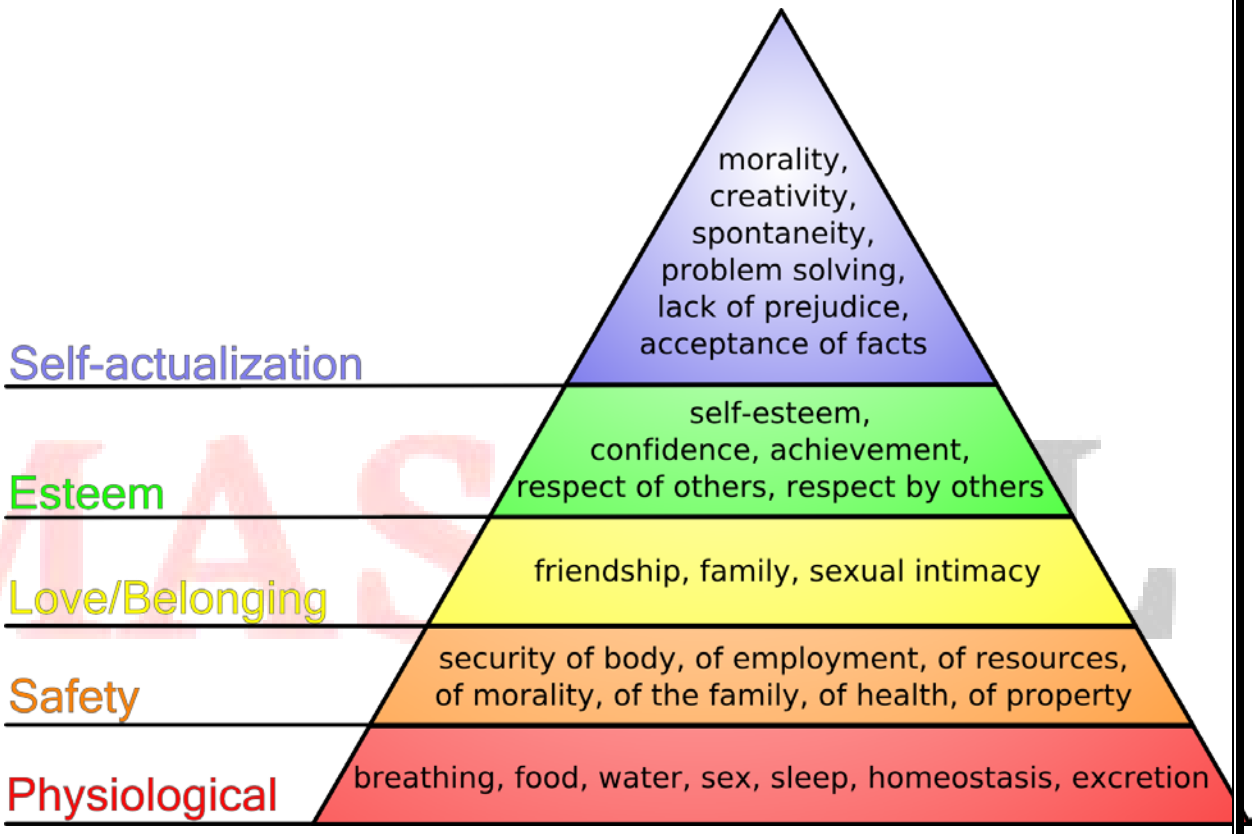
Prepotency Process Principle

- People are motivated first to satisfy the lower-order needs and then, in sequence, each of the higher order needs.
- Maslow believed that self actualization (reaching your full potential) was the ultimate psychological need.

Topic 88: Maslow's Hierarchy of Needs

1. Physiological needs
2. Safety needs
3. Affiliation and affection

4. Esteem needs
5. Self actualization



Physiological Needs

- These includes the most basic needs that are vital to survival such as the need for water, air, food and sleep.

Topic 89: Safety, Affection and Esteem Needs

Safety needs

- Much like physiological needs require maintenance throughout life, so does the need to feel secure. This need is more psychological.
- These includes needs for safety and security, e.g. a desire for steady employment , health insurance, safe neighborhoods and shelter from the environment.

Affection needs

- When physiological needs and safety needs are, by and large, taken care of a third layer starts to show up.

- You begin to feel the needs of friends, husband/wife, children, affectionate relationships in general, even a sense of community.

Esteem needs

- After the first three needs have been satisfied, esteem needs becomes increasingly important.
- These include the need for things that reflect on self esteem, personal worth, social recognition and accomplishment.

Topic 90: Self Actualization

"What a man can be he must be"

This is the highest level of Maslow's hierarchy of needs.

People are:

- Self aware
- Concerned with personal growth
- Less concerned with the opinions of others
- And interested in fulfilling their potentials

Characteristics of self actualized people:

- Acceptance and realism
- Problem centering
- Spontaneity
- Continued freshness of appreciation
- Peak experience
- The journey toward achieving a goal is just as important and enjoyable as actually accomplishing the goal.

Topic 91: Maslow's Hierarchy Applied to Education

Physiological

- Lunch time
- Correct room temperatures
- Breaks

- Sports

Safety and security

- Well planned lessons, carried out in an orderly fashion
- Controlled classroom behavior
- Emergency procedures well planned, discussed and practiced.

Lesson no 1

MASLOW'S NEED HIERARCHY II

Topic 92: Affiliation and Affection Teacher-students relationship:

- Provide positive comments and feedback rather than negative.
- Teacher personality: empathetic, considerate and interested in the individual, patient, fair, positive attitude, good listener.
- Use one-on-one instruction.
- Get to know students (like, dislikes, concern) - Be available for students in need.
- Listen to student.

Topic 93: Self esteem

- Develop new knowledge based on background knowledge so as to help to ensure success - Pace instruction to fit individual need
- Focus on strengths and assets

Respect from others

- Develop a classroom environment where students are positive and nonjudgmental.
- Develop and carry out a curriculum to encourage children to be emphatic and good listeners.
- Employ cooperative learning in such a way as to develop trust between group members.
- Involve students in activities of importance and worthiness.
- For instance, cleaning up the environment, carrying out a clothes drive for the needy.

Topic 94: Hierarchy and Knowledge

1. Physiological
2. Safety and security
3. Belongingness and affiliation
4. Esteem
5. Cognitive (1998)
6. Aesthetic
7. Self actualization
8. Self transcendence

- Allow students time to explore areas of curiosity.
- Provide lessons that are intellectually challenging.
- Plan lessons that connect areas of learning and have students compare and contrast to search for relationships
- Use a discovery approach to learning whenever possible

Topic 95: Aesthetic

- Organize classroom materials in a neat and appealing way
- Display student art work in an pleasing manner
- Put up interesting and colorful wall hangings
- Replace overly worn classroom material periodically
- Well maintained physical surroundings (e.g., keeping walls painted, desks clean and repaired etc.)
- Create varied appealing and interesting learning centers
- Rooms painted in pleasing colors
- Large window areas
- Clean rooms
- Fresh smelling rooms

Topic 96: Self Actualization

- Expect students to do their best
- Give students freedom to explore and discover on their own
- Make learning meaningful—connect to “real” life
- Plan lessons involving activities

- Get students involved in self-expressive projects
- Allow students to be involved in creative activities and projects

Lesson no 1

LEARNING STYLES I

Topic 97: Introduction

Have you ever wondered?

1. Why did you have difficulty learning from a particular teacher, whereas another seems to explain things in just the right way?
2. Did you ever questions why the course that your friend said was so easy turned into a struggle for you?

- It is based on our natural learning gifts, bents and strengths.
- It is our own individual way of inputting, processing, concentrating, remembering, understanding. Storing and processing information.
- Simply students can learn in lots of different ways known as "learning styles".

Topic 98: Growth of Learning Styles Theory 1

- In the beginning research focused on the relationship between the memory and oral or visual methods.
- In 1904 the first intelligence test was developed – generated interest in individual differences.
- The study of learning styles was next: Dr. Maria Montessori (1907), began using materials to enhance the learning styles of the students.
- Bloom's taxonomy (1956) took another step toward defining the learning style differences
- In 1984, David Kolb published his learning style model, where he determined that learning styles are closely related to cognitive skills.
- Learning styles in education recommend that teachers assess the learning styles of their students and adapt their classroom methods to best fit each student's learning style.

Topic 99: Growth of Learning Style Theory 2

Various theories on learning have been developed with increasing frequency in the last few decades:

Personality learning theories:

- Influence of basic personality on learning extraovert vs. introvert.

Information processing theories

- Individual's preferred intellectual approach to assimilating information.

Social learning theories

- Determine how students interact in the classroom.

Multidimensional and instructional theories

- Students' environmental preference for learning.

Topic 100: Basic concepts of Learning Styles

- Core concept: individual differ in how they learn.
- Assessment of learning styles typically ask people to evaluate what sort of information presentation they prefer (e.g., words versus pictures versus speech)
- And what kind of mental activity they find most engaging or agreeable (e.g. analysis versus listening)
- Instruction is best provided in a format that matches the preferences of the learner (e.g. for a "visual learner" emphasizing visual presentation of information)
- Students learning styles can change and develop and do not necessarily stay the same.
- It is possible to test children to determine their preferences

Topic 101: Importance of Learning Styles

a) Maximize success within the classroom by understanding that various learning styles and preferences exist and vary among students.

b) In corporate activities that will maximize students preference. This will make learning more enjoyable, also faster and easier for students to learn the material.

c) Creation of diverse lesson plans help:

Students to be comfortable in an optimal learning environment

Students to grow and develop beyond their learning preferences

d) Motivates the pupils and increases the teaching.

e) Stimulates the learner's learning process

f) Eases the teaching process with different aids and provides each lesson with the proper tool.

Topic 102: Learning Preferences

Learners prefer different types of input and experiences such as:

a) Some prefer structure:

They interact with the outside world when taking in information

They tend to like things decided and organized

b) Others like flexibility:

They interact with the outside world when making decisions

They like to understand and adapt to the world rather than organize it.

c) Some prefer independence while others like a social or group learning environment.

Distance education students – independent learning environment

On-campus students – working with others

d) Some prefer auditory (hearing) input, others visual (seeing) and still others like kinesthetic (movements) input

Although we have inherent preferences we will learn in most every environment.

But learning is more pleasant and lasting when our individual preferences are targeted.

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Lesson no 1

LEARNING STYLES II

Topic 103: Information Processing

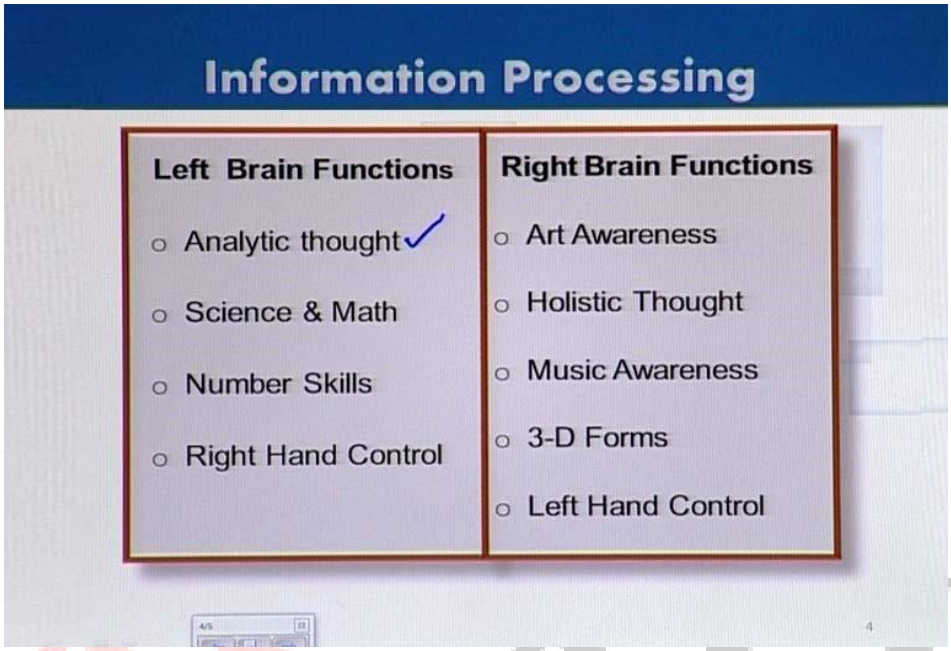
1) The first factor involved in learning styles is information processing.

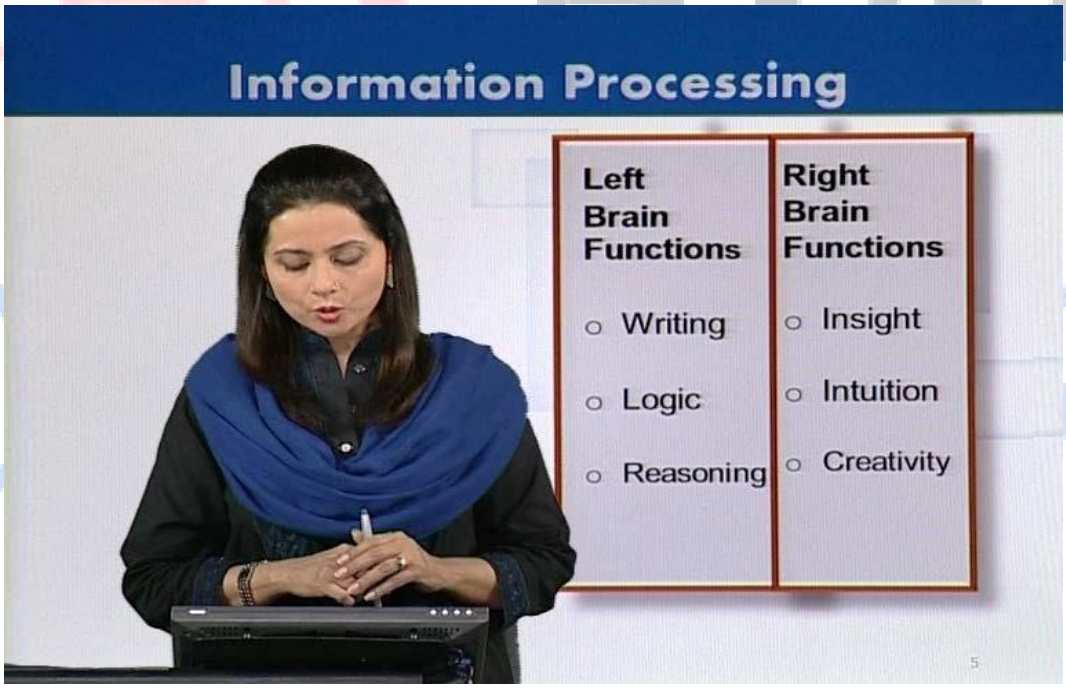
Information processing styles refer to the way in which the child concentrates, absorbs and retains information.

There are two information styles a) Analytic – prefer details

b) Global – prefer the big picture

Sometimes also called right-brain and left-brain.





Topic 104: Analytic and Global Learners

Analytic and Global Learning Theory

This theory describes the order in which a learner prefers to process information received

- By looking at the whole then breaking it down into individual parts
- Or by looking at each individual part and then combining it into a whole

Analytic learners prefer details:

- Step by step approaches, fact by fact modes
- Focused approaches, consistency
- As well as logical, objective and organized presentation of facts.

Global learners prefer seeing the broad view (the big picture):

- Using intuition
- Seeing the interrelationship between things
- Doing group activities
- Completing multiple tasks

Topic 105: Concrete and Abstract learners

The second learning style model concerns itself with the perceptual processes of knowing or cognitive abilities.

It describes the way in which we view our environment.

The two perceptual patterns are:

- Concrete
- Abstract

Concrete Learners:

Like to learn through their physical sense,

- What they can touch, see, hear, taste and smell.
- They like to deal with things that exists in the physical world.

Abstract learners:

Prefer the world of ideas and feelings.

- Abstract learners use intuitions and imaginations.

Topic 106: Sequential and Random Learners

Cognitive abilities include ordering learning styles.

- Ordering is understanding, using and storing new information in accomplishing a goal or task.

Two types of ordering styles are:

Sequential learners organize information in a step by step manner – like logical and linear thinking.

Random learners are spontaneous.

Sequential

Like clockwork, a process followed step by step, over a period of time.

Random

Like a stopwatch, starting and stopping at will, subject to what is important at the moment.

Topic 107; Learning through Perception

The next learning factor is perceptual ability.

It is the ability to be able to deal with and give meaning to sensory stimuli.

Perception – a process that involves sensing various aspects of a person, task or event and forming impressions based on selected inputs.

Three basic stages:

- Sensing characteristics
- Selecting facts
- Organizing into useful concepts

Often the way we perceive reality is colored by how we want it rather than simply the way it is.

What people perceive is usually what they believe and this is based on what they hear, see and think.

Topic 108: Perception and Reality That was quite fast!!!

Can anyone do it faster!!!

Now look at the list on the again and say the word, not the color as fast as you can!!!

Why did the second reading of the colors take longer?

The right side of the brain tries to say the color but the left side of the brain insists on readin the word.

Lesson no 1

FLEMING VAK VARK MODEL

Topic 109: Fleming's VAK VARK Model 1

- Perceptual strength relates to auditory, visual, tactual or kinesthetic learning.
- Fleming (1987) developed a list of perceptual factors designed to help students learn more about their individual learning preferences.
- Neil Fleming's VARK model is one of the most popular representations.
- Identifies 4 learning preferences:
  1. Visual
  2. Auditory
  3. Read/write
  4. Kinesthetic

Visual

Seeing: pictures, flow charts, diagrams, graphs, symbols, body language.

Likes information to be presented as charts, graphs and flow charts.

About 60% of people are visual learners.

Topic 110: Fleming's VAK VARK Model 2

Auditory

Listening/speaking: spoken words, stories, recordings, repetition, discussion, lectures, tutorials

Likes information to be presented in "Spoken word", i.e. that is "heard".

About 30% of people are auditory learners.

Read/write

Reading/writing: text, prose, essays, reports, printed words, newspapers

Likes to read about new information in all the various forms, lecture, notes, books, articles, web pages etc.

Kinesthetic

Move/doing: smells, tastes, case studies, multisensory experiences, role-play, lab sessions.

Learners best by doing the tasks – stimulated o0r real

About 10% of the general population are kinesthetic learners.

Topic 111: Learning Styles

Advantages and Disadvantages

The visual style

Advantage: makes recollection easier when in an environment which is different from where you had learned the information.

Disadvantage: the difficulty when only texts and speeches are available for learning, without any visual aids.

The auditory style

Advantage: you assimilate and retain information without having to see it in texts or pictures.

Disadvantages: difficulty of learning among silently reading learners – for example, in a liberary.

The read/write style

Advantage: makes pupil more self dependent because they can learn much by themselves.

Disadvantage: not being able to learn easily where the only medium of instruction is visual or audio; or lack of access to writing materials.

The kinesthetic style

Advantage: of exposing learners faster to practice and evidence: you practice and practice what you learn

Disadvantage: where there are no places to move to for such live experience and nobody to interact with.

Topic 112; Benefits of Knowing your LS

It is beneficial to find out:

The learning style that suits you

To know what kind of learner you are

Some of these benefits include:

Academic advantages

- Maximize your learning potential - Succeed on all educational levels
- Understand how to best study and score better on exams and tests

Personal advantages

- Improve your self confidence and self esteem
- Learn how to best use your brain
- Gain insight into your own strengths as well as weaknesses
- Learn how to enjoy learning more

Student benefits

- Know how they learn and make smarter choices
- Target areas that need improvement

Educator benefits

- Develop better lesson plans
- Give all students an equal opportunity to succeed **Topic 113: What is your learning style?**

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Topic 114: Critique

- Those who oppose the use of learning styles put forward a number of objections.
- Many of these are in fact criticisms of the way that learning styles are misused rather than a criticism of the concept itself
- It is argued that using learning styles could be detrimental because:

  Should not be used as mere labels

  Rather it should appreciate full range of factors
- Teachers should not focus exclusively on one style
- Interconnectivity should be appreciated
- Children may be discouraged from trying activities that do not fit with their own preference
- Teachers may think a child's preferred style is as unchangeable, forgetting that they are just acquired habits.

Lecture no 2

KOLB MODEL

Topic 115: Kolb's Learning Cycle

Introduction

"Learning is the process whereby knowledge is created through the transformation of experience"

(David A. Kolb 1984, 38)

- Kolb's (1984) theory sets out four distinct learning styles (or preferences) which are based on a four stage learning cycle.
- Kolb's model offers both a way to understand individual learning styles, and also an explanation of a cycle of experiential learning.
- Kolb proposed that an individual learner moves through a spiral of immediate experience which leads to observations and reflections on the experience
- These reflections are then absorbed and linked with previous knowledge and translated into abstract concepts or theories
- Which results in new ways and actions to adjust to the experience that can be tested and explored

Topic 117: Reflective Observation

- In this stage of the learning cycle, people understand ideas and situations from different points of view.
- In a learning situation the learner would rely on patience, objectivity and careful judgment but would not necessarily take any action.
- The learners would rely on their own thoughts and feelings in forming opinions.
- In the example, after finishing tutorial, the students reflects on what they did, makes observations and discusses how they went with their teacher.

Topic 118: Abstract Conceptualization

- In this stage, learning involves using theories, logic and ideas rather than feelings, to understand problems or situations.
- Typically, the learner relies on systematic planning and develops theories and ideas to solve problems
- In the example, the student then thinks about the tutorial process and their performance and tries to make links between previous experience of learning.

Topic 119: Active Experimentation

- Learning in this stage takes an active form – experimenting with changing situations.
- The learner would take a practical approach and be concerned with what really works, as opposed to simply watching a situation.
- In the example the student considers ways to improve, and tries out methods and strategies based on the previous stages of the cycle.

Topic 120

Implications of Kolb Model



- The cycle can be entered at any point but all stages must be followed in sequence for successful learning to take place
- The learning cycle suggests that it is not sufficient to have an experience in order to learn.
- It is necessary to reflect on the experience to make generalizations and formulate concepts.
- Which can then be applied to new situations.
- This learning must then be tested out in new situation.
- The learner must make the link between the theory and action by planning, acting out, reflecting and relating it back to the theory.

Lesson no 2

KOLB LEARNING STYLES

Topic 121: Collaborative and Cooperative learning

Introduction

Teaching vs Learning

John Amos Comenius, a 16th Century scholars; summarized the approach that teaching should follow, "the main object is to find a method by which teachers teach less but learners learn more", proving that current problems have noble pedigrees.

Definition

An instruction method in which learners work in groups towards a common academic goal.

"collaborative learning is an umbrella term for a variety of educational approaches involving joint intellectual effort by students, or students and teachers together".

(Smith and McGregor: 1992)

Topic 122: Kolb's Learning Styles

Explanation of Learning Styles

- Kolb said that learning is a combination of:
- How we approach a task and how we respond to and assimilate the experience.

In approaching a task (processing) we have a preference for

- either doing
- or watching

In responding to the experience we have a preference for

- either feeling
- or thinking

The combination of these preferences creates four main learning styles

For example, a person with a dominant learning style of "doing" rather than watching the task and "feeling" rather than "thinking" about the experience will have a learning style which combines these processes, namely an "accommodating" learning style in Kolb's terminology.

Topic 123: Learning Styles Continuum

Knowing your and your student's learning style enables learning to be positioned with the preferred method.

Experience can be tailored to best fit with the given situation and a student's learning style preference.

Topic 124: Characteristics of Learning Styles

Accommodator-CE/AE (feeling and doing)

- Is a hands-on learner
- Likes to take new challenges
- Is a team worker relies more on intuition than logic
- Is target oriented depends on others for problem solving

Diverger-CE/RO

- Likes to watch more than do
- Is sensitive, emotional and creative
- Prefers group work
- Uses own imagination for problem solving
- Shows open mind and accepts personal feedback

Assimilator-AC/RO

(thinking and watching)

- A thinker; needs time to think
- Prefers a to the point, logical approach to learning
- Likes to have clear explanation instead of practical opportunity
- Creates abstract ideas and theories

Converge-AC/AE

(thinking and doing)

- Technical minded, likes technical tasks
- Likes experimentation, test new ideas
- Applies learning to find solutions to practical situations

Topic 125: Argument for Learning Styles

- People with a distinct learning style preferences will be likely to achieve more success in their learning when the process of learning is tuned with their preference.
- **For example** assimilators without their notes and clear instructions will be very uncomfortable in such a learning situation.
- Accommodators will be easily frustrated if they are made to read a lot of instruction and rules and are not allowed to get hands-on experience as soon as possible.
- **For example** differences in learning styles of teachers and their students can also lead to development of tension between them.

Topic 126: Argument Against Learning Styles

- Process of reflection: not adequate
- Claims about the four learning styles: exaggerated
- Cultural conditions/experiences: not addressed adequately
- The idea of stages: does not match reality empirical evidence: weak
- Learning styles theories: largely unproven methodologies counter-productive in same cases of questionable benefits
- Recognizing that students have different preferred learning styles, the fact needs to be built in to the design of learning activities and assessment

Conclusion

Kolb's learning cycle provides a useful framework for designing effective active learning experiences

DISCOVERY LEARNING

Topic 127: Eureka Effect

- Hiero II asked Archimedes to find a method for determining whether a crown was pure gold or mixed with silver.
- While taking a bath in bath tub he realized that a given weight of silver would displace more water than an equal weight of gold.
- At this point the shouted, "EUREKA" (I have found it!)
- Discovery learning is based on this "Aha!" method.
- This Aha! Moment occurs when a learner finally comes across the answer to a problem on their own and when things suddenly begin to make sense.
- --- although in some of out classrooms, "Aha!" sounds more like, Aaaaaahhhhh!

Topic 128: Inquiry-Based Learning

- Discovery learning is an inquiry based learning method, and is based on the **constructivist** school of thought.
- Discovery learning takes place in **"problem solving situations".**
- In a problem solving situation, learners draw on their own past experience and current knowledge to discover new truths and facts and thus they learn.
- In this way, students are more likely to remember concepts and knowledge when it is discovered by themselves than in transmission model which assumes that learning is mainly dependent on the teacher.

"You can not teach people everything they need to know. The best you can do is position them where they can find what they need to know when they need to know it".

(Seymour Papert)

Topic 129: Discovery Learning in Practice

Using discovery learning for obtaining **educational goals** will include:

- Promoting a deep understanding.
- Developing **meta- cognitive** skills.
- Encouraging a high level of student engagement.

It is process of **"inductive inquiry"**- learners conducting experiments like in scientific process:

1. Identify variables

2. Collect data
3. Interpret data
4. Make conclusions about information

Discovery learning has three main **characteristics:**

1. Exploration and problem solving;
2. Student cantered activities based on student interest ; and
3. Scaffolding new information into students' funds of knowledge.

Topic 130: Forms of Discovery Learning 1

Experiment – a systematic procedure undertaken with the purpose of

- Confirming, or
- Contradicting

The validity of a theory

Even very young children perform basic experiments to learn about the world.

Exploration – to look for information and resources.

Simulation- based Learning – provides learners with an experience of working on simplified model of reality reflecting some or all of its properties – e.g. medical education.

Topic 131:Forms of Discovery Learning 2

Inquiry-based Learning – where knowledge is built from experience and process, especially socially based experience.

- Learning develops best in group situations.

Problem-based Learning – provides learners with an experience of working on simplified model of reality reflecting some or all of its properties – e.g. medical education.

Webquests – an inquiry oriented lesson format in which most of all the information that learners work with comes from the web.

- These can be created using a simple word processing document that includes links to websites.

Topic 132: Levels of Inquiry based Learning

Confirmation Inquiry

- Learner is given the problem and the procedure (method).
- Results are known in advance.
- Confirmation of the results is the object of the inquiry.

Useful to reinforce a previously learned idea.

Structured inquiry

- Learner is given the problem and the procedure (method).
- Task is to generate an explanation that is supported by the evidence collected in the procedure.

Students evaluate and analyze the collected data.

Guided inquiry

- Learner is given the problem only (research question).
- Task is to design the procedure (method).
- And to test the question and the resulting explanations.

More open than the previous two.

Open inquiry

- Learner formulates questions.
- Designs procedures (method) for carrying out an inquiry. - Communicates results.

Often seen in science fair contexts where students pursue their own investigative questions.

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